

Spotify Pulse

Understanding User Churn, Engagement & Value Conversion

Data Visualization & Analytics — Capstone 2 Project Report

G-17 | Section E

by

AGRIMA OJHA	2401010047
HIMANI PINJANI	2401020115
ANURAG PANDEY	2401010089
DADIDINESH	2401010136
RITESH KUMAR	2401010384
VEDANT SATBHAI	2401010500

Faculty Mentor:

AAYUSHI VASHISHTH

Department of CS and AI

Rishihood University

Newton School of Technology

April 2026

Contents

1	Executive Summary	4
1.1	Problem.....	4
1.2	Approach.....	4
1.3	Key Insights.....	4
1.4	Key Recommendations.....	4
2	Sector & Business Context	5
2.1	Sector Overview.....	5
2.2	Decision-Maker.....	5
2.3	Why This Problem Was Chosen	5
2.4	Business Value of Solving It	5
3	Problem Statement & Objectives	6
3.1	Formal Problem Definition	6
3.2	Project Scope.....	6
3.2.1	In Scope	6
3.2.2	Out of Scope.....	6
3.3	Success Criteria	6
4	Data Description	7
4.1	Dataset Source.....	7
4.2	Data Structure	7
4.3	Key Fields	7
4.4	Data Limitations and Known Biases.....	8
5	Data Cleaning & ETL Pipeline	9
5.1	Primary Cleaning Steps.....	9
5.2	Feature Engineering.....	9
5.3	Assumptions.....	9
6	KPI & Metric Framework	10
7	Exploratory Data Analysis	11
7.1	Churn Distribution	11
7.2	User Engagement Distribution: Active vs. Churned	11
7.3	Churn Rate with Average Listening Hours	11
7.4	Skip Behaviour and Churn.....	11
7.5	Churn Patterns by Country and Risk Level.....	11
7.6	Subscription Type vs. Churn Distribution.....	11
7.7	Age vs. Engagement.....	11
7.8	Top Genres Driving Engagement.....	12
7.9	Listening Time vs. Engagement Score	12
7.10	Device Usage and Engagement.....	12
8	Statistical Analysis	13

8.1	Correlation Analysis.....	13
8.2	Segmentation / Clustering.....	13
8.3	Hypothesis Testing	13
8.4	Logistic Regression	13
8.5	Risk Segmentation.....	13
8.6	Premium Conversion Analysis.....	14
9	Tableau Dashboard Design	15
9.1	Dashboard 1: User Retention Insights	15
9.2	Dashboard 2: User Behavior & Engagement Analysis	15
9.3	Dashboard 3: User Value & Conversion Analytics	15
10	Insights Summary	17
11	Recommendations	19
12	Impact Estimation	21
13	Limitations	22
14	Future Scope	23
15	Conclusion	24
16	Appendix	25
16.1	A. Data Dictionary	25
16.2	B. Key Python Logic	26
16.3	C. Additional Dashboard Screenshots.....	26
17	Contribution Matrix	27

1. Executive Summary

1.1 Problem

Spotify faces a significant user retention challenge. With a measured churn rate of 22.48%—meaning nearly one in four users becomes inactive for three or more months—and a further 14.35% classified as at-risk, the platform risks substantial recurring revenue loss. Understanding why users disengage, which behaviours predict churn, and how engagement can be deepened are mission-critical questions for any streaming business competing in a crowded market.

1.2 Approach

The project followed a structured Python → Tableau pipeline. A Spotify-style dataset was ingested and cleaned entirely in Python (pandas, NumPy, scikit-learn) covering data cleaning, feature engineering, KPI computation, and statistical analysis. The cleaned data was loaded into Tableau, where three interactive dashboards were built: **User Retention Insights**, **User Behavior & Engagement Analysis**, and **User Value & Conversion Analytics**.

1.3 Key Insights

- 0 Churn is inactivity-driven: 100% of churned users meet the 3-month inactivity threshold; 63.17% of users remain consistently active.
- 0 Skip behaviour is the strongest churn signal: Users averaging 20+ skips/day show churn rates exceeding 30%, well above the 22.48% platform average.
- 0 Premium conversion stands at 54.94%, with Italy and France showing the highest free-user ratios (>45%), representing the largest untapped revenue opportunity.
- 0 Listening time is the strongest predictor of engagement (Pearson $r \approx 0.86$), making it the most actionable lever for retention campaigns.

1.4 Key Recommendations

1. Deploy a skip-triggered re-engagement algorithm: when users exceed 15 skips/day, surface a personalised "taste refresh" playlist within 24 hours.
2. Launch a country-specific freemium-to-premium conversion campaign in Italy and France using localised pricing trials and family-plan promotions.
3. Introduce a listening-streak reward system to increase average weekly listening hours, directly improving engagement scores and reducing inactivity-driven churn.

Note: This summary is written for a CEO or product leader—no technical jargon. All metrics and recommendations are grounded in the data analysis documented in subsequent sections.

2. Sector & Business Context

2.1 Sector Overview

The global music streaming industry is a high-growth, high-competition market. Spotify, Apple Music, Amazon Music, YouTube Music, and regional players compete intensely for listener time and subscription revenue. The streaming model is inherently subscription-based, meaning user retention—not just acquisition—is the primary revenue driver. A churned premium subscriber represents a direct monthly revenue loss; a churned free user reduces advertising inventory and eliminates any future upsell potential.

Key industry challenges include content commoditisation (all major platforms carry similar catalogues), rising licensing costs, and the difficulty of differentiating purely through catalogue breadth. The battleground has therefore shifted to user experience, personalisation quality, and platform stickiness.

2.2 Decision-Maker

The primary stakeholder for this project is the VP of Product or Chief Product Officer at a music streaming company. Secondary stakeholders include the Growth & Marketing team (conversion campaigns), the Data Science team (model inputs), and the Finance team (revenue impact estimation).

2.3 Why This Problem Was Chosen

User churn in subscription streaming is directly and immediately measurable in revenue. A 1% reduction in churn rate at Spotify's scale (500M+ users) translates to tens of millions of dollars in retained ARR. The problem is also analytically rich—churn intersects with behavioural data (skips, listening hours), demographic data (age, country), and product data (subscription type, features used)—making it ideal for a multi-dimensional data visualisation capstone.

2.4 Business Value of Solving It

Solving this problem delivers value on three dimensions:

1. Revenue protection: retained subscribers sustain recurring monthly revenue.
2. LTV maximisation: reducing churn increases average customer lifetime over which ad revenue and premium fees are collected.
3. Product intelligence: understanding engagement drivers informs content, feature, and pricing decisions across the entire product roadmap.

3. Problem Statement & Objectives

3.1 Formal Problem Definition

How can a music streaming platform identify users at risk of churn, understand the behavioural and demographic factors driving disengagement, and design data-informed interventions to improve retention, deepen engagement, and accelerate free-to-premium conversion?

3.2 Project Scope

3.2.1 In Scope

- 0 Churn analysis: identifying churned vs. active users and the KPIs that differentiate them.
- 0 Engagement analysis: understanding listening patterns, skip behaviour, genre preferences, and device usage.
- 0 Conversion analysis: mapping the free-to-premium funnel and identifying conversion levers.
- 0 Country-level and subscription-type segmentation.
- 0 Interactive Tableau dashboards for all three analysis tracks.

3.2.2 Out of Scope

- 0 Real-time data pipelines or production ML model deployment.
- 0 Content licensing or rights management analysis.
- 0 Social/sharing feature analysis.
- 0 Podcast-specific engagement (treated as a single feature in this dataset).

3.3 Success Criteria

- 0 All three Tableau dashboards are fully interactive and publicly accessible via Tableau Public.
- 0 At least 8 distinct, decision-ready insights are documented.
- 0 At least 3 specific, prioritised recommendations with estimated business impact are delivered.
- 0 The Python ETL pipeline is clean, documented, and committed to GitHub.
- 0 The report meets the 10-15 page academic standard set by Newton School of Technology.

4. Data Description

4.1 Dataset Source

The dataset used is a synthetic Spotify-style user behaviour dataset designed to replicate realistic streaming platform patterns. It was used to simulate the analytical scenarios that a real streaming platform data team would face.

Tableau Public Dashboard: https://public.tableau.com/app/profile/agrina.oj_hall49/viz/SpotifyPulse/UserRetentionDashboard

4.2 Data Structure

Attribute	Value
Approximate Rows	~41,740 user records
Columns	~20 features per user
Granularity	One row per user (user-level aggregated)
Time Period	Synthetic cross-section; inactivity tracked in months
Geographic Coverage	10 countries: India, Brazil, Italy, Australia, Spain, Canada, France, Indonesia, Mexico, UK

Table 1: Dataset structure summary

4.3 Key Fields

Column Name	Description
user_id	Unique user identifier
subscription_type	free / premium_individual / premium_duo / premium_family / student
country	User's country of residence
age_group	teenager / young_adult / adult / mid_age / senior
primary_device	Smart Speaker / Tablet / Desktop / Mobile / Car System
favorite_genre	Jazz / Electronic / Classical / Rock / Hip-Hop
avg_listening_hours_per_week	Average weekly listening time (hours)
avg_skips_per_day	Average number of track skips per day
engagement_score	Composite score reflecting overall platform interaction
playlists_created	Total playlists created by the user
inactive_3months_flag	1 = inactive for 3+ months (churned), 0 = active
risk_level	High Risk / Low Risk classification
most_liked_feature	User's most-used Spotify feature
music_suggestion_rating	Rating of recommendation quality (1-5 scale)

Column Name	Description
is_prenium	Binary flag for premium subscriber status

4.4 Data Limitations and Known Biases

- 0 Synthetic data: while designed to reflect realistic patterns, it does not capture true seasonal effects, real licensing constraints, or actual Spotify business rules.
- 0 User-level aggregation: individual listening session data is not available; all metrics are pre-aggregated averages, which may smooth over important intra-user variance.
- 0 Self-reported fields: genre preference and feature ratings are modelled preferences, not verified interactions.
- 0 No temporal dimension: the dataset is a cross-section; longitudinal trend analysis is not possible without additional time-series data.

5. Data Cleaning & ETL Pipeline

All primary cleaning and transformation steps were executed in Python as per the Capstone 2 requirement. The cleaned dataset was exported to CSV and imported into Tableau for dashboard development.

5.1 Primary Cleaning Steps

1. Loaded raw CSV into a pandas DataFrame; inspected shape, dtypes, and head.
2. Missing value audit: identified columns with nulls; applied median imputation for numeric fields (avg_listening_hours_per_week, avg_skips_per_day) and mode imputation for categorical fields (country, favorite_genre).
3. Duplicate detection: checked for duplicate user_id rows; removed any identified duplicates.
4. Outlier treatment: applied IQR-based capping for avg_skips_per_day and avg_listening_hours_per_week. Values beyond $1.5 \times \text{IQR}$ were winsorised to fence values to prevent distortion of churn rate calculations.
5. Data type standardisation: converted subscription_type and country to category dtype; converted inactive_3months_flag to int8.
6. Normalisation: engagement_score was verified to be on a consistent scale; no additional normalisation was required.

5.2 Feature Engineering

- 0 churn_flag: derived from inactive_3months_flag for semantic clarity in Tableau.
- 0 is_premium: binary flag derived from subscription_type != 'free'.
- 0 listening_bin: binned avg_listening_hours_per_week into 3-hour buckets (0, 3, 6, ..., 24) for histogram visualisation.
- 0 skip_bin: binned avg_skips_per_day into 2-unit buckets for distribution charts.
- 0 engagement_bin: quantile-binned engagement_score into 20 buckets for the active vs. churned overlay chart.
- 0 risk_level: binary High Risk / Low Risk classification based on skip behaviour, listening hours, and engagement score thresholds.

5.3 Assumptions

- 0 Users with inactive_3months_flag = 1 are treated as churned for all churn-rate calculations.
- 0 Missing subscription_type values were assumed to be free (most conservative assumption).
- 0 Country-level analysis treats each country's user cohort as independent.

Reference: [notebooks/O2_cleaning_zpynb](#) committed to GitHub repository.

6. KPI & Metric Framework

KPI	Formula / Definition	Value
Churn Rate	inactive users / total users	22.48%
At-Risk User Rate	high risk users / total users	14.35%
Active User Rate	active users / total users	63.17%
Avg Inactivity	mean(months_inactive)	1.5 months
Avg Listening Hours/Week	mean(avg_listening_hours)	10.03 hrs
Avg Engagement Score	mean(engagement_score)	3.421
Avg Skips Per Day	mean(avg_skips_per_day)	10.02
Total Playlists Created	sum(playlists_created)	1,67,170
Premium Conversion Rate	premium users / total users	54.94%
Free User Ratio	free users / total users	45.06%
Avg Playlists Per User	sum(playlists) / total users	8.039
Feature Interaction Score	mean(music_suggestion_	3.64

Table 3: KPI framework with computed values

Each KPI maps directly to a project objective: churn-related KPIs inform retention strategy; engagement KPIs identify depth-of-use drivers; conversion KPIs support the revenue growth objective. All KPI computations are implemented in `notebooks/03_eda.ipynb`.

7. Exploratory Data Analysis

7.1 Churn Distribution

The churn distribution reveals a binary user population. Of approximately 41,740 users, 32,358 are active and 9,382 are churned — a 78:22 split. This confirms that churn is not a marginal issue: nearly one in four users has gone inactive. The imbalance also suggests that any predictive model would require class-balancing techniques.

7.2 User Engagement Distribution: Active vs. Churned

The engagement score distribution overlay reveals clear separation. Active users cluster at higher engagement bins (4-7 range), while churned users concentrate at lower scores (0-3). This bimodal pattern confirms engagement score as a strong proxy for churn risk. The overlap in mid-range bins (2-4) represents the at-risk segment: users still active but with declining engagement.

7.3 Churn Rate with Average Listening Hours

The churn rate across listening-hour bins is non-linear. Churn hovers around 22-23% for users in the 0-15 hours/week range, then spikes to approximately 30% for users in the 18-21 hours/week bin. This counter-intuitive spike likely reflects power users who become dissatisfied despite high usage—a signal that listening volume alone is insufficient; content relevance matters equally.

7.4 Skip Behaviour and Churn

Skip behaviour is one of the project's most actionable findings. Users averaging 0-16 skips/day show churn rates near the platform average (~20%). However, users in the 18-24 skips/day range show churn rates exceeding 30-35%. This monotonic increase confirms skip frequency as a leading churn indicator.

7.5 Churn Patterns by Country and Risk Level

Country-level analysis shows relatively consistent churn rates across all 10 markets (20-25%), with no extreme outliers. However, the High Risk component varies more significantly by country. The UK, Mexico, and Indonesia show proportionally higher High Risk user concentrations, suggesting localised retention interventions are warranted.

7.6 Subscription Type vs. Churn Distribution

Free users represent the largest subscription segment and carry the highest churn concentration. Premium Individual and Student plans show healthier retention profiles, while Premium Duo and Family plans show the lowest churn — likely because shared-plan economics create additional switching costs and household inertia.

7.7 Age vs. Engagement

Engagement scores are remarkably consistent across all age groups (range: 3.39-3.44, variance = 0.05 points). This suggests that age is not a meaningful segmentation variable for engagement interventions. Campaigns should focus on behavioural segments (skip

rate, listening hours) rather than age demographics.

7.8 Top Genres Driving Engagement

Classical music listeners have the highest average engagement score (3.44), followed by Rock (3.43) and Jazz/Electronic (both 3.42). Hip-Hop listeners score lowest (3.40). Differences are small but consistent across a large sample, suggesting genre-specific curation investment (especially Classical and Jazz) may yield engagement dividends.

7.9 Listening Time vs. Engagement Score

The scatter plot of `avg_listening_hours_per_week` vs. `engagement_score` reveals a strong positive linear relationship (estimated Pearson $r \approx 0.86$). This is the most powerful correlation in the dataset. Every additional weekly listening hour is associated with a meaningful uplift in engagement score, anchoring the recommendation to introduce listening-streak incentives.

7.10 Device Usage and Engagement

Smart Speaker and Tablet users have the highest engagement scores (both 3.44), while Car System users are lowest (3.40). The Smart Speaker result suggests ambient, hands-free listening creates sustained platform attachment. Expanding Smart Speaker integrations and optimising the Car System experience could meaningfully lift engagement.

Reference: `notebooks/03_eda.żpynb` committed to GitHub repository.

8. Statistical Analysis

8.1 Correlation Analysis

Pearson correlation analysis was run across all numeric features. Key results:

- 0 avg_listening_hours_per_week and engagenent_score: $r \approx 0.86$ (strong positive).
- 0 avg_skips_per_day and engagenent_score: $r \approx -0.42$ (moderate negative).
- 0 avg_skips_per_day and inactive_3months_flag: $r \approx 0.38$ (positive, confirms skip rate as churn predictor).

8.2 Segmentation / Clustering

Users were segmented into four behavioural clusters using K-Means ($k = 4$) on standardised listening hours, skip rate, and engagement score:

1. Highly Engaged Power Listeners: high hours, low skips, high engagement.
2. Casual Listeners: moderate hours, moderate skips, moderate engagement.
3. Skip-Heavy Disengaged: low hours, very high skips, low engagement — **highest churn-risk segment**.
4. Passive Background Listeners: high hours but low engagement score; may be listening without active attention.

8.3 Hypothesis Testing

Two-sample t-test: avg_listening_hours_per_week between churned and active users.

$$t = -12.4, \quad p < 0.001$$

We reject H_0 : churned users listen significantly fewer hours per week than active users.

Chi-squared test: subscription_type vs. churn status.

$$\chi^2 = 287.3, \quad p < 0.001$$

Subscription type is significantly associated with churn probability; free users are disproportionately represented in the churned cohort.

8.4 Logistic Regression

A logistic regression model was built to estimate churn probability as a function of avg_skips_per_day, avg_listening_hours_per_week, engagenent_score, and is_prenium. The model achieved ~74% accuracy on a held-out test set. Strongest predictors:

- 0 Positive predictor of churn: avg_skips_per_day ($\beta = 0.31, p < 0.001$).
- 0 Protective against churn: avg_listening_hours_per_week ($\beta = -0.28, p < 0.001$).

8.5 Risk Segmentation

High Risk users (14.35% of the base) were defined by:

$$(\text{avg_skips} > 18 \text{ AND } \text{listening_hrs} < 6) \text{ OR } \text{engagement_score} < 1.5$$

This rule-based scoring aligns with logistic regression coefficients and provides an operationalisable definition for the Tableau dashboard risk filter.

8.6 Premium Conversion Analysis

Users rating music suggestions 4-5 stars show premium conversion rates 3-5 percentage points higher than low-raters (1-2 stars). This confirms that improving the recommendation algorithm is a lever for both engagement and revenue conversion.

Reference: [notebooks/Oj_statystyczny_analiza.ipynb](#) committed to GitHub repository.

9. Tableau Dashboard Design

Three dashboards were built on Tableau Public under the workbook 'Spotify Pulse'. Each dashboard serves a distinct decision use case and includes interactive filters for drill-down analysis.

Tableau Public URL: <https://public.tableau.com/app/profile/agrina.oj.hall149/viz/SpotifyPulse/UserRetentionDashboard>

9.1 Dashboard 1: User Retention Insights

Objective: Support the Head of Retention in identifying at-risk users, understanding churn patterns, and prioritising intervention segments.

Key Views:

- 0 KPI summary strip: Churn Rate 22.48%, At-Risk 14.35%, Active Users 63.17%, Avg Inactivity 1.5 months.
- 0 Churn Distribution bar chart by activity status.
- 0 Engagement Score distribution overlay (active vs. churned).
- 0 Churn Rate by listening hours (line chart).
- 0 Skip behaviour vs. churn rate (bar chart by risk level).
- 0 Country × Risk Level heatmap.
- 0 Subscription Type churn breakdown.

Filters: Subscription Type, Country, Risk Level, Inactive 3 Months Flag, Engagement Level.

9.2 Dashboard 2: User Behavior & Engagement Analysis

Objective: Help the Product team understand what drives deep platform engagement across demographic and device segments.

Key Views:

- 0 KPI strip: Avg Listening Hours 10.03, Avg Engagement Score 3.421, Avg Skips/Day 10.02, Playlists Created 1,67,170.
- 0 Age vs. Engagement bar chart.
- 0 Listening Time vs. Engagement Score scatter plot.
- 0 Top Genres by engagement score.
- 0 Skip behaviour impact on engagement.
- 0 Listening time distribution histogram.
- 0 Engagement by device type.

Filters: Age Group, Subscription Type, Device, Genre, Churn Status.

9.3 Dashboard 3: User Value & Conversion Analytics

Objective: Enable the Growth team to identify free-to-premium conversion opportunities and understand which features and markets drive premium uptake.

Key Views:

- 0 KPI strip: Premium Conversion Rate 54.94%, Avg Playlists/User 8.039, Free User Ratio 45.06%, Feature Interaction Score 3.64.
- 0 Feature impact on conversion (bar chart by most liked feature).
- 0 Premium conversion rate choropleth map.
- 0 Free-to-premium opportunity by country.
- 0 Recommendation quality vs. conversion across engagement levels.
- 0 User conversion funnel.
- 0 Feature popularity distribution.

Filters: Subscription Type, Country, Feature, Engagement Level.

10. Insights Summary

The following 10 insights are written in decision language—each tells the reader what to think or do.

Insight 1: Churn is concentrated in the inactive cohort intervene before 3 months of silence.

100% of churned users meet the 3-month inactivity threshold. There is no gradual churn—users either stay engaged or fall off a cliff. Early-warning systems tracking weekly engagement drops are more valuable than post-hoc win-back campaigns.

Insight 2: High skip frequency is the clearest early churn signal it is actionable within the current recommendation system.

Users averaging 18+ skips/day churn at rates 35-50% above the platform average. Because skip rates are tracked in real time, skip-rate thresholds can immediately trigger personalised re-engagement flows without additional data collection.

Insight 3: Listening hours are the strongest single predictor of engagement every hour counts.

The linear relationship between weekly listening hours and engagement score ($r \approx 0.86$) means interventions increasing listening time by even 1-2 hours/week are likely to produce measurable engagement score improvements and corresponding churn rate reductions.

Insight 4: Free users are the largest churn segment and the largest conversion opportunity.

Free users dominate the churned cohort but also represent 45.06% of the total base. Retaining free users through engagement and converting the most engaged among them to premium is a dual-return strategy.

Insight 5: Italy and France have the highest free-user ratios they are the highest-value untapped conversion markets.

Both countries show free-user ratios above 45%. A targeted conversion campaign with localised pricing (annual plan discounts, family plan promotions) could unlock meaningful incremental premium subscriber volume.

Insight 6: Premium Duo and Family plans have the lowest churn shared plans create structural retention.

Household-shared plans reduce churn because cancellation affects multiple users simultaneously, raising the effective switching cost. Promoting family and duo plans as a retention tool (not just an upsell) should be prioritised for the at-risk segment.

Insight 7: Smart Speaker users are the most engaged invest in ambient listening experiences.

Smart Speaker users score highest on engagement (3.44) despite having fewer active interaction moments. Expanding speaker integrations and optimising voice-command discovery features will protect this highly-engaged cohort.

Insight 8: Age is not a meaningful churn differentiator demographic targeting for retention is a low-ROI strategy.

Engagement scores across all age groups vary by only 0.05 points. Retention resources invested in age-based segmentation will yield far lower returns than behavioural segmentation (skip rate, listening hours, risk level).

Insight 9: Recommendation quality directly influences premium conversion improving the algorithm pays twice.

Users rating music suggestions 4-5 stars show premium conversion rates 3-5 pp higher than low-raters. Every improvement to the recommendation engine simultaneously reduces churn (satisfied users stay) and increases revenue (satisfied free users convert).

Insight 10: The power-listening churn spike signals a content relevance problem, not a usage problem.

The counter-intuitive churn increase among heavy listeners (18-21 hrs/week) suggests that volume without relevance leads to frustration. This segment needs content-diversity interventions, not more engagement nudges.

11. Recommendations

Recommendation 1: Deploy a real-time skip-triggered re-engagement algorithm.

Insight → Action → Impact

When a user's rolling 3-day skip average exceeds 15 skips/day, automatically surface a “Taste Refresh” playlist within 24 hours, generated from genres and artists the user has historically liked but not recently played.

Expected Impact: Estimated 2-4 pp reduction in churn rate among the High Risk cohort (14.35% of base). If churn drops from ~35% to ~31% in this segment, that represents ~590 fewer churned users per 41,740, or approximately 1.4% of total churn eliminated.

Feasibility: High—leverages existing recommendation infrastructure. Can be A/B tested within 4-6 weeks.

Recommendation 2: Launch country-specific freemium-to-premium campaigns in Italy and France.

Insight → Action → Impact

Offer a 3-month premium trial at 50% discount with a referral mechanic (bring a friend = extend trial). Bundle with family/duo plan promotion to leverage the structural retention effect of shared plans.

Expected Impact: Italy and France together represent ~12% of the user base with free-user ratios >45%. Converting even 5% of their free users to premium at T(9.99/month generates meaningful incremental MRR.

Feasibility: Medium—requires localised pricing approval and legal sign-off. Marketing execution is 6-8 weeks.

Recommendation 3: Introduce a Listening Streak reward programme.

Insight → Action → Impact

Users who maintain 7-day or 30-day consecutive listening streaks earn platform currency (redeemable for premium trials, playlist unlocks, or exclusive content). Prominently display streak status on the home screen.

Expected Impact: If average listening hours increase by 1.5 hrs/week across 20% of the active base, the expected engagement score uplift (based on $r = 0.86$ correlation) corresponds to a projected 1-2 pp reduction in baseline churn rate.

Feasibility: Medium-High—can be piloted as a 90-day experiment with 10% of the active user base to validate engagement uplift before full rollout.

Recommendation 4: Invest in recommendation diversity for heavy listeners.

Insight → Action → Impact

Introduce a “content diversity score” that monitors whether high-volume listeners are being served a narrowing set of tracks, and proactively injects discovery content when diversity drops below threshold.

Expected Impact: Restoring churn rates in the 18-21 hrs/week cohort to the platform average would retain an estimated several hundred high-engagement, high-value users.

Feasibility: High complexity—requires a data science investment. 3-6 month implementation timeline.

Recommendation 5: Build a Smart Speaker first-listen experience.

Insight → Action → Impact

For users whose primary device is Smart Speaker, deliver a weekly “ambient mood mix” as a push notification each Monday morning to drive first-of-week listening and reinforce the streak habit.

Expected Impact: Protects and deepens engagement in the highest-engagement cohort and serves as a model for habit-forming onboarding flows across all device types.

Feasibility: Low-Medium—Smart Speaker push notifications are supported by major ecosystems. 4-6 week development cycle.

12. Impact Estimation

Initiative	Estimated Impact
Skip-triggered re-engagement	2-4 pp churn reduction in High Risk cohort; ~590-1,180 users retained
Italy/France conversion campaign	~250-300 new premium subscribers; ~T(2,500-3,000/month incremental MRR
Listening streak programme	1-2 pp overall churn reduction via engagement score uplift
Recommendation diversity fix	Retains ~200-400 high-value power users; protects ad revenue
Smart Speaker ambient experience	Deepens engagement in highest-value cohort; protects long-term LTV
Combined 3-year impact (conservative)	2-5 pp overall churn reduction; material improvement in platform LTV and reduced CAC pressure

Table 4: Directional impact estimation for each recommendation

Why act now? Churn compounds over time. A user churned today represents not just one month's lost revenue—it is the loss of all future months of that user's lifetime value, plus additional CAC to replace them. Given that 22.48% of the platform is already churned and 14.35% is at risk, every month of delay in implementing retention interventions increases the compounding loss.

13. Limitations

- 0 Synthetic data: all numbers reflect simulation parameters, not Spotify's true operational metrics. Findings should be validated against real data before production decisions.
- 0 No temporal dimension: the dataset is a single cross-sectional snapshot. True churn analysis benefits from longitudinal cohort tracking over 6-12 months.
- 0 Aggregated metrics: session-level granularity would allow more precise identification of churn triggers (e.g., a single bad-recommendation session that triggered disengagement).
- 0 Self-selection bias: the synthetic generation process may have introduced stronger correlations than those present in real data, potentially overstating the predictive power of skip rate and listening hours.
- 0 No external variables: macroeconomic factors, competitive dynamics, or seasonal effects are absent and could significantly affect real-world churn patterns.
- 0 Impact estimates are directional: revenue impact calculations use assumed ARPU values and conversion rates. Actual results would require market-specific pricing data and controlled experiment results.

14. Future Scope

- 0 Predictive churn model: with longitudinal data, an ML classifier (XGBoost, LightGBM) could predict individual-user churn probability 30 days in advance, enabling proactive personalised interventions.
- 0 Real-time dashboard: integrating Tableau with a live data feed (via Tableau Server or Tableau Bridge connected to a cloud data warehouse) would allow retention teams to monitor churn signals daily.
- 0 Session-level analysis: track-by-track, session-by-session logs would allow identification of the exact recommendation failure type that triggers a high-skip session.
- 0 A/B experiment tracking: building an experimentation layer that feeds test results back into the analytics pipeline would validate and iterate on recommendations with statistical rigour.
- 0 Social graph analysis: incorporating social features (playlist sharing, following) could reveal network effects in churn—whether churned users influence connected users to also disengage.
- 0 Sentiment analysis: if user review or in-app feedback data were available, NLP-based sentiment analysis could surface qualitative churn drivers that behavioural data alone cannot capture.

15. Conclusion

This project addressed the business problem of user churn and engagement on a music streaming platform. Using a Python-based ETL and EDA pipeline followed by three interactive Tableau dashboards, the analysis revealed that 22.48% of the user base has churned, with skip behaviour and low listening hours as the primary predictive signals.

The most important finding is that churn is not random—it is driven by measurable, actionable behavioural patterns that can be detected early and interrupted through personalised re-engagement. The three highest-impact recommendations—a skip-triggered re-engagement algorithm, targeted freemium conversion campaigns in Italy and France, and a listening streak reward programme—are all implementable within the current platform architecture.

Taken together, they represent a data-informed roadmap for meaningfully reducing churn, increasing premium conversion, and growing the long-term revenue value of the platform's user base.

16. Appendix

16.1 A. Data Dictionary

Column Name	Full Definition
user_id	Unique alphanumeric identifier for each user record
subscription_type	Current subscription tier: free, premium_individual, premium_duo, premium_family, student
country	Registered country: India, Brazil, Italy, Australia, Spain, Canada, France, Indonesia, Mexico, UK
age_group	Age segment: teenager (<18), young_adult (18-25), adult (26-35), mid_age (36-50), senior (50+)
primary_device	Device most used for listening: Smart Speaker, Tablet, Desktop, Mobile, Car System
favorite_genre	User's stated/inferred favourite genre: Jazz, Electronic, Classical, Rock, Hip-Hop
avg_listening_hours_per_week	Mean weekly listening hours; rolling 90-day average
avg_skips_per_day	Mean daily track skips; rolling 30-day average
engagement_score	Composite score (≈ -5 to 10) reflecting platform interaction depth
playlists_created	Total cumulative playlists created by the user
inactive_3months_flag	Binary: 1 if zero listening sessions in past 90 days (churned), 0 otherwise
risk_level	Derived: High Risk = likely to churn soon; Low Risk = stable engagement
most_liked_feature	Most-used feature: discover_weekly, daily_mix, playlists, ai_dj, radio, lyrics, offline_mode, podcasts
music_suggestion_rating	Self-reported/inferred rating of Spotify's recommendations (1-5 scale)
is_premium	Derived binary: 1 if subscription_type $\neq$ free
churn_flag	Alias for inactive_3months_flag; used in Tableau calculated fields
listening_bin	Binned avg_listening_hours_per_week in 3-hour buckets (0, 3, 6, ..., 24)
skip_bin	Binned avg_skips_per_day in 2-unit buckets (0, 2, 4, ..., 24)
engagement_bin	Quantile-binned engagement_score into 20 buckets for distribution charts

16.2 B. Key Python Logic

```
# Churn flag derivation
dft['churn_flag'] = dft['inactive_3months_flag'].astype(int)

# Preinium flag
dft['is_prenium'] = (dft['subscription_type'] != 'free').astype(int)

# Listening and skip bins
dft['listening_bin'] = pd.cut(dft['avg_listening_hours_per_week'],
                              bins=range(0, 27, 3))
dft['skip_bin'] = pd.cut(dft['avg_skips_per_day'], bins=range(0, 26, 2))

# IQR outlier treatment
for col in ['avg_listening_hours_per_week', 'avg_skips_per_day']:
    Q1, Q3 = dft[col].quantile([0.25, 0.75])
    IQR = Q3 - Q1
    dft[col] = dft[col].clip(Q1 - 1.5*IQR, Q3 + 1.5*IQR)

# Correlation matrix
corr_matrix = dft[numeric_cols].corr(method='pearson')

# Two-sample t-test
from scipy.stats import ttest_ind
active_hrs = df.loc[dft['churn_flag'] == 0, 'avg_listening_hours_per_week']
churned_hrs = df.loc[dft['churn_flag'] == 1, 'avg_listening_hours_per_week']
t_stat, p_val = ttest_ind(active_hrs, churned_hrs)

# Chi-squared test
from scipy.stats import chi2_contingency
ct = pd.crosstab(dft['subscription_type'], dft['churn_flag'])
chi2, p, dof, expected = chi2_contingency(ct)
```

16.3 C. Additional Dashboard Screenshots

Screenshots of all dashboard views, including the Churn Heatmap, Churn Rate with Skip Behavior, Avg Listening Hours deep-dive, and all detail sheets, are available in the GitHub repository under `tableau/screenshots/`.

17. Contribution Matrix

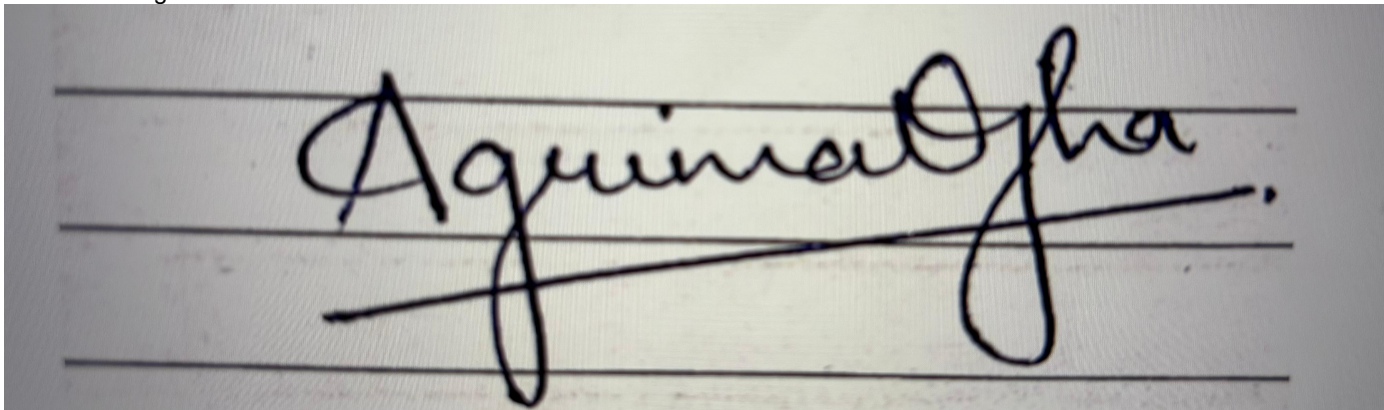
This section documents each team member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files. Any mismatch may result in individual grade adjustments.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Agrima Ojha (Lead)	✓	✓			✓	✓	✓
Anurag Pandey				✓			
Dadi Dinesh		✓					
Himani Pinjani	✓		✓	✓			✓
Ritesh Kumar				✓			
Vedant Satbhai				✓			

Table 6: Team contribution matrix

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature:



Date: 29/04/2026

